Can Traditional Volatility Models Explain Bitcoin?
A Comparative GARCH Analysis of Bitcoin and the S&P 500

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I. Introduction

This study examines whether Bitcoin exhibits similar volatility patterns to the S&P 500 and whether stock market volatility can help explain and forecast Bitcoin volatility. Bitcoin is often viewed as a high-risk digital asset, while the S&P 500 represents broad U.S. equity market performance. By comparing their volatility behavior over the same period, this study evaluates whether traditional volatility models, such as GARCH, can be applied to cryptocurrency markets.

II. Objective

The objective of this study is to examine whether Bitcoin follows similar volatility dynamics to traditional stock markets and whether stock market volatility can help forecast Bitcoin volatility.

III. Research Questions

1. Relative Volatility: Is Bitcoin significantly more volatile than the S&P 500?
2. Statistical Behavior: Do both assets exhibit volatility clustering and fat tails?
3. Predictability: Can S&P 500 volatility help explain or forecast Bitcoin volatility?

IV. Methodology

The study compares volatility levels, examines statistical properties, and evaluates predictive relationships to model the volatility dynamics of Bitcoin and the S&P 500. A detailed description of the methodology is presented in *Appendix B.*

Data Description

Daily price data for Bitcoin (*BTC-USD*) and the S&P 500 (*^GSPC*) are obtained from Yahoo Finance for January 2020 to December 2025. To ensure comparability, both series are aligned on common trading dates, resulting in 1507 observations. Daily log returns (in percentage terms) are computed and used for all analyses.

Descriptive Statistics

Summary statistics, including mean, standard deviation, skewness, kurtosis, and extreme values, are used to assess volatility, tail behavior, and return distributions.

ARCH Test

An ARCH test is conducted to detect volatility clustering.

- (H_0): No ARCH effects
- (H_1): ARCH effects exist

Rejection of H_0 ($p < 0.05$) supports the use of GARCH-type models.

GARCH(1,1)

Volatility is modeled using the *GARCH(1,1)* framework:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where α captures the impact of shocks and β reflects persistence. Volatility persistence is measured by α + β, with values close to 1 indicating slow decay. To validate the model specification, alternative GARCH models, including *GARCH(1,2), (2,1),* and *(2,2),* are also tested, with model selection based on AIC and BIC.

GJR-GARCH Model

To capture asymmetric effects, the *GJR-GARCH(1,1)* model is used:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \gamma I_{\{\epsilon_{t-1}<0\}}\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where γ measures the impact of negative shocks. A significant γ indicates asymmetric volatility (leverage effects).

Model Evaluation

Models are compared using AIC and BIC, with lower values indicating better fit. Ljung–Box tests on residuals and squared residuals are used to assess remaining autocorrelation and ARCH effects.

Forecasting Bitcoin Using Stock Volatility

To examine whether S&P 500 volatility helps explain Bitcoin volatility, an *EGARCH* model with an external regressor is estimated by incorporating lagged stock market volatility in the variance equation:

$$\sigma_{BTC,t}^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \gamma\sigma_{SP,t-1}^2$$

where $\sigma_{BTC,t}^2$ denotes the conditional variance of Bitcoin returns, $\sigma_{SP,t-1}^2$ represents the lagged volatility of the S&P 500, and γ captures the spillover effect from stock market volatility. A statistically significant γ coefficient indicates that stock market volatility has explanatory power for Bitcoin volatility, suggesting the presence of volatility transmission from traditional financial markets to cryptocurrency markets.

V. Results and Interpretation/Analysis

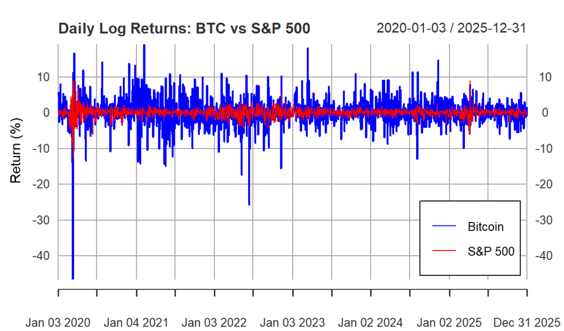


Figure 1. Daily Log Returns: BTC vs S&P 500

Figure 1 illustrates that Bitcoin returns exhibit substantially larger fluctuations than those of the S&P 500 throughout the sample period. Bitcoin displays more frequent and more extreme positive and negative return spikes, providing clear evidence of higher short-term volatility and greater sensitivity to market shocks.

The return patterns also suggest clear volatility clustering. Periods of large Bitcoin movements are often followed by further large movements, while calmer periods tend to persist. This behavior is consistent with time-varying volatility and supports the use of ARCH/GARCH-type models in subsequent analysis. Although the S&P 500 also exhibits volatility clustering, its return fluctuations are more tightly centered around zero and are less extreme in magnitude.

From a statistical perspective, the wider dispersion of Bitcoin returns indicates higher unconditional volatility and stronger tail risk. The presence of large negative spikes suggests that Bitcoin is particularly exposed to downside shocks, which may be driven by liquidity constraints, regulatory developments, macroeconomic uncertainty, or shifts in investor sentiment.

Summary Statistics

The descriptive statistics (*see Appendix C - Table 1*) further support these observations. Bitcoin exhibits a standard deviation of approximately 3.93, significantly higher than the 1.32 observed for the S&P 500, confirming that Bitcoin is considerably more volatile on a day-to-day basis. Bitcoin also demonstrates more pronounced extreme movements, with returns ranging from approximately −46 to +19, whereas the S&P 500 displays a much narrower range, reflecting lower exposure to large price swings.

In addition, both assets exhibit negative skewness and high kurtosis, indicating that their return distributions deviate from normality and are characterized by fat tails. These features are more pronounced for Bitcoin, suggesting higher tail risk and a greater likelihood of extreme market events. The high kurtosis further indicates a leptokurtic distribution, consistent with the presence of large and infrequent shocks.

Overall, Bitcoin exhibits higher volatility, more extreme return behavior, and stronger fat-tailed characteristics than the S&P 500, while both series display volatility clustering, supporting the use of GARCH-type models.

Conditional Volatility Interpretation

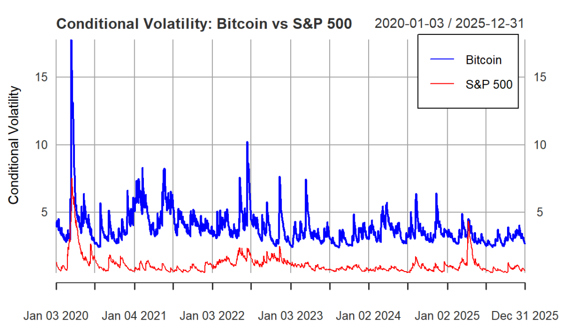


Figure 2. Conditional Volatility: Bitcoin vs S&P 500

For conditional volatility, the estimated time-varying volatility of Bitcoin and the S&P 500 is compared. The results, *Figure 2*, show that Bitcoin maintains consistently higher volatility levels throughout the sample period, providing clear evidence of greater sensitivity to market news, investor sentiment, and liquidity shocks. Bitcoin also exhibits more frequent and pronounced volatility spikes, particularly during periods of market stress, reflecting stronger reactions to sudden events and heightened uncertainty.

In contrast, the S&P 500 demonstrates lower and smoother volatility dynamics, indicating greater market stability and more gradual adjustments to macroeconomic developments. These findings imply that Bitcoin not only has higher average volatility but also more unstable and persistent volatility over time.

Furthermore, both series display clear volatility clustering, supporting the application of GARCH-type models for volatility estimation. The observed co-movement in volatility between Bitcoin and the S&P

500 suggests exposure to common market shocks and provides preliminary evidence of potential volatility transmission across the two markets. *(see Appendix L)*

Forecast Performance and Validation

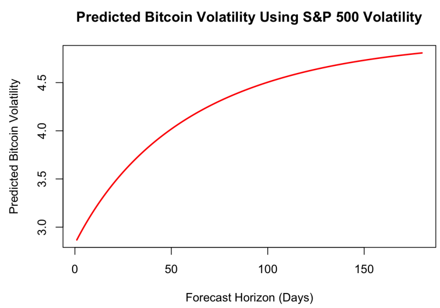


Figure 3. Predicted Bitcoin Volatility Using S&P 500 Volatility

Figure 3 presents the forecasted Bitcoin volatility using both the baseline GARCH model and the extended model incorporating S&P 500 volatility. The two forecast paths are closely aligned, with the S&P-augmented model producing a marginally higher volatility trajectory. This indicates that while stock market volatility contributes some predictive information, its incremental impact on Bitcoin volatility remains limited.

To assess the reliability of these forecasts, we compare them with observed market behavior from January 1 to April 30, 2026 (*see Appendix H & I - Figures 4 and 5*). During this period, Bitcoin continues to exhibit substantially higher realized volatility than the S&P 500, with frequent large return fluctuations, while the S&P 500 remains comparatively stable. This confirms that the model accurately captures the relative volatility dynamics between the two assets.

In terms of magnitude, the forecasted volatility paths align well with the general level and persistence observed in the data. For example, Bitcoin's forecast converges toward a higher long-run volatility level (~4.5–4.8), which is consistent with realized volatility levels observed during early 2026, where large price swings remain frequent. Similarly, the S&P 500 forecast stabilizes at a lower level (~1.3–1.4), reflecting its more stable return behavior.

However, the model exhibits some smoothing behavior and does not fully capture short-term extreme volatility spikes, particularly for Bitcoin. This suggests that while the GARCH framework effectively models volatility persistence and long-run behavior, it may underestimate tail risk in highly speculative markets.

The results indicate that the model provides reliable forecasts for relative volatility levels and long-run dynamics. While incorporating S&P 500 volatility adds some predictive value, its contribution remains limited, suggesting that Bitcoin volatility is primarily driven by asset-specific and speculative factors rather than traditional market influences. Consequently, forecasts beyond May 2026 are most informative for capturing general trends and baseline risk levels, rather than short-term volatility fluctuations.

VI. Conclusion

This study examines whether Bitcoin exhibits similar volatility dynamics to the S&P 500 and whether stock market volatility can help explain and forecast Bitcoin volatility. The results show that Bitcoin is substantially more volatile than the S&P 500, with larger return fluctuations, greater tail risk, and stronger sensitivity to market shocks.

Both assets exhibit key stylized facts of financial time series, including volatility clustering, persistence, and fat-tailed distributions. However, these characteristics are more pronounced in Bitcoin, indicating less stable and more reactive return dynamics compared to traditional equity markets.

The GARCH modeling results provide further insight into these differences. Based on model selection criteria using AIC and BIC, the GARCH(1,1) specification is identified as the most suitable model for capturing the volatility dynamics of both assets. It effectively models persistence and time-varying volatility, although it is less successful in capturing extreme short-term fluctuations, particularly for Bitcoin. *(see Appendix L)*

The extended model incorporating S&P 500 volatility shows that stock market volatility contains some predictive information for Bitcoin. However, its incremental contribution is limited, suggesting that Bitcoin volatility is primarily driven by asset-specific and speculative factors rather than traditional market influences.

Overall, while Bitcoin shares certain statistical properties with traditional financial assets, its volatility behavior remains fundamentally different. Traditional volatility models provide a useful baseline but are not sufficient on their own to fully capture the dynamics of cryptocurrency markets.

Real World Implications

The results indicate that while Bitcoin may offer diversification benefits, it carries significantly higher volatility and tail risk compared to traditional equities. Standard risk management approaches based on equity markets may underestimate cryptocurrency risk, particularly during periods of stress. In addition, the observed co-movement in volatility suggests that diversification benefits may weaken during market turbulence, highlighting the need for robust risk management and careful asset allocation when incorporating Bitcoin into portfolios. *(see Appendix N)*

Limitations

This study has several limitations. The use of daily data may not fully capture intraday volatility dynamics in continuously traded cryptocurrency markets. In addition, standard GARCH-type models may not fully reflect the complexity of Bitcoin volatility, as indicated by remaining ARCH effects. The analysis is also limited to a single cryptocurrency and equity index and does not incorporate macroeconomic or crypto-specific factors, which may affect the generalizability of the results. *(see Appendix N)*

Further Research

Future research can extend this analysis by applying more advanced volatility models and incorporating additional explanatory variables, including macroeconomic and crypto-specific factors. Expanding the scope to multiple assets and using high-frequency data may further improve the understanding and forecasting of cryptocurrency volatility dynamics. *(see Appendix N)*

Appendix

A. R Code

https://drive.google.com/drive/folders/1ZUxuZ2CZueFEW6Qt2jXmf2UCo3qxuzQp?usp=sharing

B. Methodology

The methodology compares volatility levels, examines statistical properties, and evaluates predictive relationships using a range of quantitative techniques to model the volatility dynamics of Bitcoin and the S&P 500.

Data Description
The dataset consists of daily price data for Bitcoin (BTC-USD) and the S&P 500 index (^GSPC), obtained from Yahoo Finance, covering the period from January 2020 to December 2025. Bitcoin trades continuously, including weekends, while the S&P 500 follows standard trading days. To ensure comparability, both series are aligned based on common trading dates. After cleaning and alignment, the final dataset consists of 1507 observations. Daily log returns are then computed from price data and used for all statistical analysis.

Return Structure
Returns are computed as the logarithmic differences of prices, which provide time-additive and scale-invariant properties. To facilitate interpretation, returns are multiplied by 100 and expressed in percentage terms.

Descriptive Statistics
To compare the volatility patterns of Bitcoin and the S&P 500, descriptive statistics are computed, including the mean, standard deviation, skewness, kurtosis, and the minimum and maximum returns. These measures provide insights into relative volatility, the presence of fat tails, and the frequency of extreme price movements.

ARCH Test
Before applying GARCH models, the ARCH test is used to detect volatility clustering.

Hypothesis:
(H_0): No ARCH effects
(H_1): ARCH effects exist

If the p-value is less than 0.05, the null hypothesis is rejected, indicating volatility clustering and supporting the use of GARCH-type models.

GARCH(1,1)
The conditional variance of returns is modeled using the GARCH(1,1) framework, which captures volatility clustering commonly observed in financial time series. The standard GARCH(1,1) specification is defined as follows:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where r_t denotes the return at time t, σt^2 represents the conditional variance. The parameters α and β measure the impact of past shocks and past volatility, respectively, reflecting the persistence of volatility over time. Volatility persistence is measured by the sum α+β; a value close to 1 indicates that volatility shocks are highly persistent and decay slowly over time.
To ensure model robustness, alternative GARCH specifications are also tested, including GARCH(1,2), GARCH(2,1), and GARCH(2,2). These higher-order models allow for additional lag structures in shocks and volatility. Model selection is based on information criteria such as AIC and BIC, enabling comparison of model fit and determination of the most appropriate specification for each asset.

GJR-GARCH Model

To account for asymmetric effects, where negative shocks may exert a stronger influence on volatility than positive shocks of equal magnitude, the GJR-GARCH(1,1) model is employed. The specification is given by:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \gamma I_{\{\epsilon_{t-1}<0\}}\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where γ captures the asymmetry effect, and $I_{\{\epsilon_{t-1}<0\}}$ is an indicator variable that equals 1 when the lagged shock is negative and 0 otherwise. A statistically significant γ coefficient indicates that negative shocks have a larger impact on volatility than positive shocks of the same magnitude. This asymmetry reflects the presence of leverage effects, where adverse news leads to greater increases in volatility compared to favorable news, suggesting that the standard GARCH model may be insufficient to fully capture the volatility dynamics.

Model Comparison

To determine the best-fitting model, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are employed, with lower values indicating superior model performance. In addition, Ljung–Box tests are applied to both standardized residuals and squared residuals to assess the presence of remaining autocorrelation and ARCH effects, thereby ensuring the adequacy of the model specification.

Forecasting Bitcoin Using Stock Volatility

To examine whether S&P 500 volatility helps explain Bitcoin volatility, an EGARCH model with an external regressor is estimated by incorporating lagged stock market volatility in the variance equation:

$$\sigma_{BTC,t}^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \gamma\sigma_{SP,t-1}^2$$

where $\sigma_{BTC,t}^2$ denotes the conditional variance of Bitcoin returns, $\sigma_{SP,t-1}^2$ represents the lagged volatility of the S&P 500, and γ captures the spillover effect from stock market volatility. A statistically significant γ coefficient indicates that stock market volatility has explanatory power for Bitcoin volatility, suggesting the presence of volatility transmission from traditional financial markets to cryptocurrency markets.

C. Table 1. Comparative Descriptive Statistics: Bitcoin vs. S&P 500 Returns

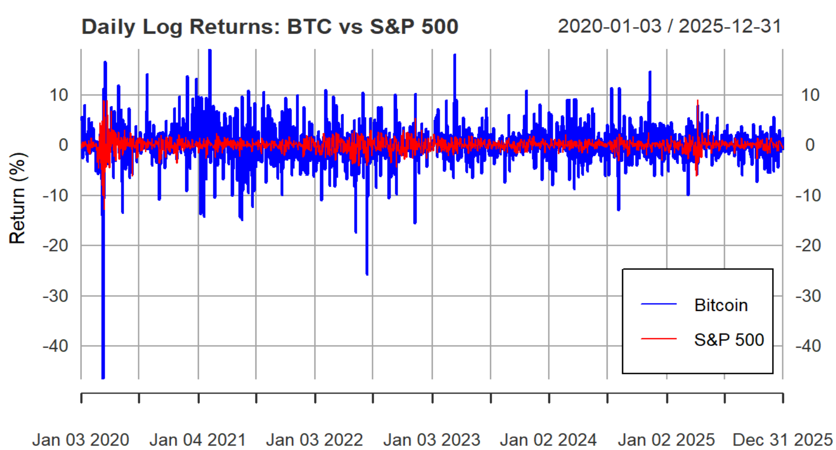
Asset	Mean	SD	Skewness	Kurtosis	Min	Max
Bitcoin	0.16774435	3.926458	-1.2216037	19.29263	-46.47302	19.152671
S&P500	0.04927167	1.322136	-0.6448749	17.62961	-12.76522	9.089488

D. Table 2. Comparison of GARCH Orders

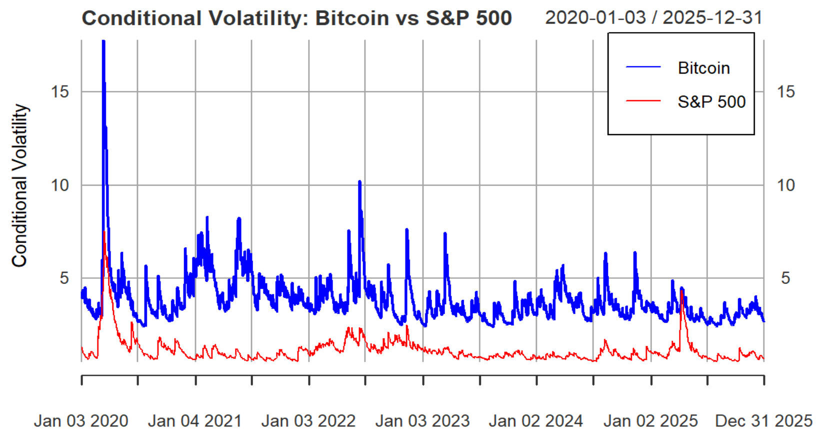
	(1,1)		**(1,2)**		**(2,1)**		**(2,2)**	
	BTC	**S&P**	**BTC**	**S&P**	**BTC**	**S&P**	**BTC**	**S&P**
Akaike	5.262814	2.834286	5.264188	2.835934	5.264188	2.834874	5.265515	2.836201
Bayes	5.280458	2.851930	5.285360	2.857106	5.285360	2.856047	5.290216	2.860902
Shibata	5.262792	2.834264	5.264156	2.835902	5.264156	2.834842	5.265472	2.836158
Hannan-Quinn	5.269385	2.840857	5.272073	2.8438197	5.272073	2.842760	5.274715	2.845401

The information criteria for the GARCH(1,2) and GARCH(2,1) specifications are nearly identical for Bitcoin, indicating that both models capture similar volatility dynamics. This suggests that Bitcoin volatility is highly persistent, and the model can flexibly distribute this persistence between shock terms and lagged variance terms without significantly improving model fit. As a result, increasing model complexity does not provide additional explanatory power, supporting the use of a more parsimonious GARCH(1,1) specification.

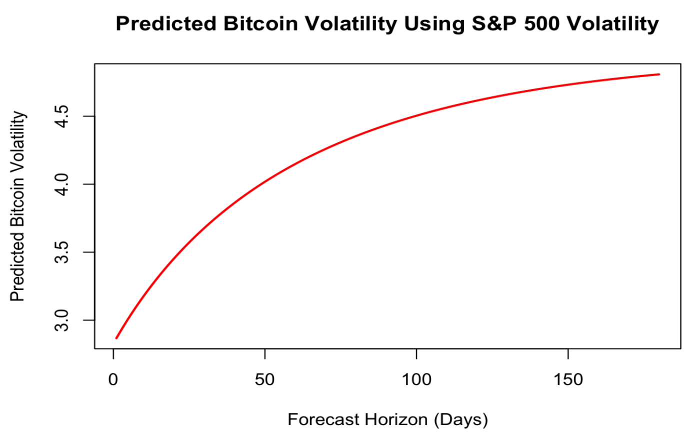
E. *Figure 1. Daily Log Returns: BTC vs S&P 500*



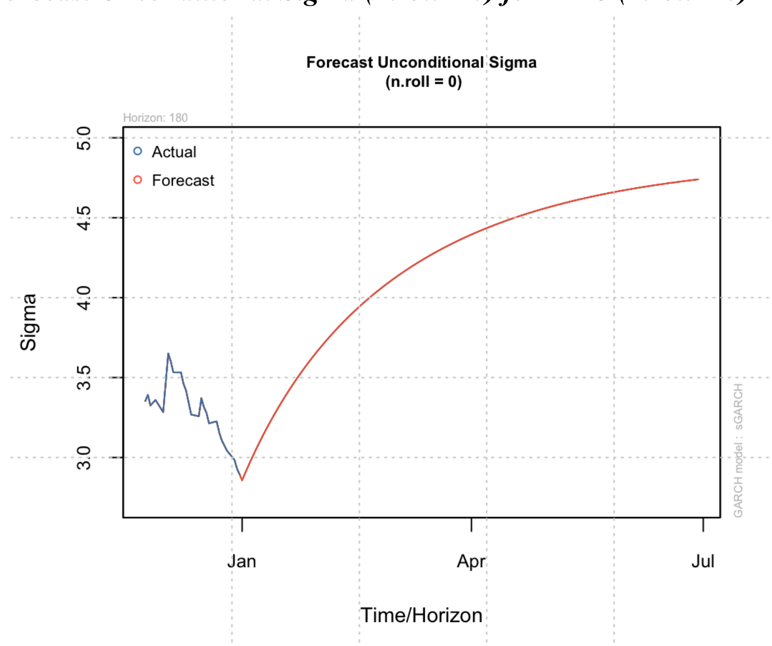
F. *Figure 2. Conditional Volatility: Bitcoin vs S&P 500*



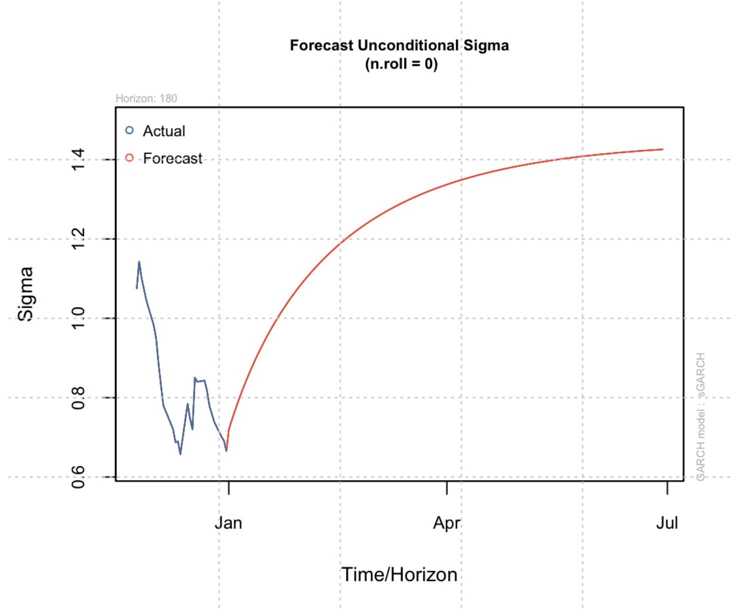
G. *Figure 3. Predicted Bitcoin Volatility Using S&P 500 Volatility*



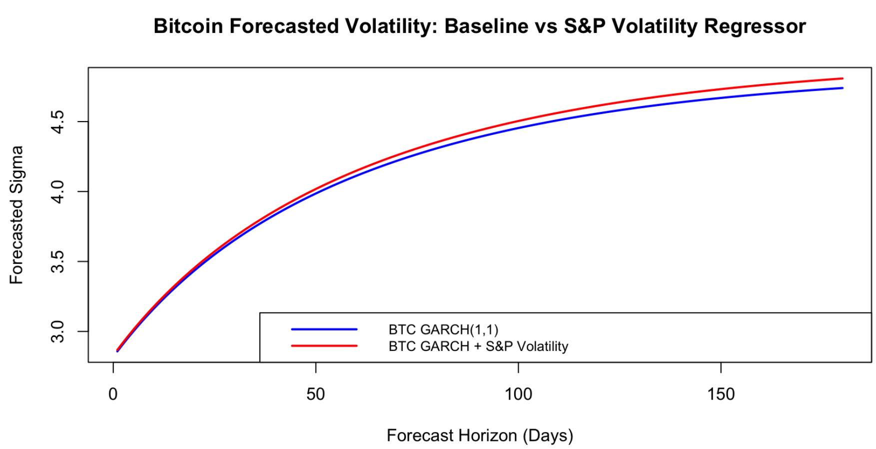
H. *Figure 4. Forecast Unconditional Sigma (n.roll = 0) for BTC (n.roll = 0)*



I. *Figure 5. Forecast Unconditional Sigma (n.roll = 0) for S&P 500 (n.roll = 0)*



J. Figure 6. Bitcoin Forecasted Volatility: Baseline vs S&P Volatility Regressor



K. ARCH Test - Null hypothesis: No ARCH effects

Data	Model	Chi-squared	df	p-value	Result
Bitcoin	ARCH	32.214	12	0.001282	Reject Null
S&P500	ARCH	492.06	12	< 2.2e-16	Reject Null

L. AIC/BIC Model Comparison

Data	Model	Akaike	Bayes	Shibata	Hannan-Quinn
Bitcoin	GARCH(1,1)	5.262814	5.280458	5.262792	5.269385
S&P500	GARCH(1,1)	2.834286	2.851930	2.834264	2.840857
Bitcoin	GJR-GARCH	5.263934	5.285107	5.263903	5.271820
S&P500	GJR-GARCH	2.802596	2.823769	2.802565	2.810482
Bitcoin	GARCH with S&P Volatility Regressor	5.262940	5.284124	5.262908	5.270830

M. Ljung-Box Test - Null hypothesis

Data	X-squared	df	p-value	Result
BTC Residuals	15.143	10	0.1269	Reject Null
BTC Squared Residuals	20.123	10	0.02811	Fail to Reject Null
S&P Residuals	9.6267	10	0.4738	Reject Null
S&P Squared Residuals	6.1882	10	0.7992	Reject Null

** No autocorrelation in the data up to the specified number of lags

N. Real World Implications, Limitations, and Further Research

Real World Implications

The findings carry important implications for investors, risk management, and portfolio construction. Bitcoin offers potential diversification benefits due to its distinct return and volatility drivers; however, these benefits come with significantly higher volatility and pronounced tail risk. As a result, Bitcoin cannot be treated as a direct substitute for traditional equities, as its risk-return profile differs fundamentally from that of the S&P 500.

From a risk management perspective, standard approaches such as Value-at-Risk (VaR) and volatility-based risk measures must be adjusted to account for Bitcoin's persistent and clustered volatility. Models calibrated on equity market behavior may underestimate the magnitude and persistence of risk in cryptocurrency markets, particularly during periods of stress.

Furthermore, while Bitcoin may contribute to diversification under normal market conditions, the observed co-movement in volatility suggests that common shocks and spillover effects can emerge during periods of market turbulence. This reduces diversification benefits precisely when they are most needed. Overall, these results highlight the importance of robust risk management frameworks and careful asset allocation when incorporating Bitcoin into investment portfolios.

Limitations

Despite the insights provided, this study is subject to several limitations. The analysis relies on daily data, which may not fully capture intraday volatility dynamics that are particularly relevant in cryptocurrency markets, where trading occurs continuously. The use of higher-frequency data could reveal additional patterns in volatility behavior.

In addition, the study employs relatively standard GARCH-type models, which may not fully capture the complexity of Bitcoin volatility. The presence of remaining ARCH effects in the residuals suggests that richer model specifications may be required to better describe its dynamics.

The scope of the analysis is also limited to a single cryptocurrency and a single equity index, which may restrict the generalizability of the findings to other digital assets or financial markets. Finally, the models do not explicitly incorporate macroeconomic variables, regulatory developments, or crypto-specific factors such as ETF flows, all of which may play a significant role in driving Bitcoin volatility.

Further Research

Future research can extend this analysis in several directions to better capture the complexity of cryptocurrency markets. More advanced volatility models, such as regime-switching GARCH frameworks or models with heavy-tailed error distributions, may provide a more accurate representation of Bitcoin's volatility dynamics.

Incorporating additional explanatory variables, including macroeconomic indicators, interest rates, liquidity conditions, and crypto-specific factors, could further improve the predictive power of volatility models. Expanding the analysis to include multiple cryptocurrencies and financial assets would also provide a broader perspective on cross-market volatility relationships and potential spillover effects.

Finally, the use of high-frequency data may offer deeper insights into intraday volatility patterns and enhance the accuracy of short-term volatility forecasting, particularly in markets characterized by continuous trading and rapid information flow.